Literature Review

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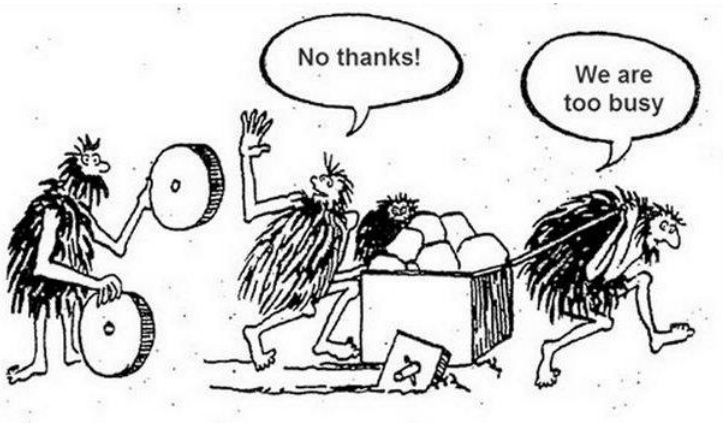
Why do you need a literature review?

- There are 2 main reasons for writing a literature review.
  - To justify and guide a research project.
  - To create original and valuable research work.
    - **Alert: new knowledge is produced!**

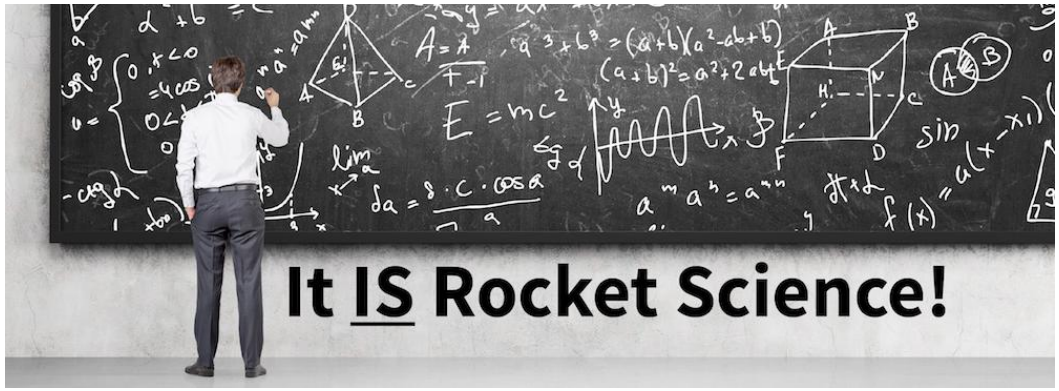
A justification and a guide for a research plan

- Identifying what has been **published or posted** on a **subject or topic.**
  - What are the **questions** and **problems**?
  - How **relevant** are the questions and problems?
    - How often are the questions mentioned?
    - What happens if a good answer is found?
  - What **answers** and **solutions** have been tried?
    - Where and why did others fail or succeed in quests for answers?
  - **Background**.
    - What are the theoretical foundations?
    - What do I need to know?

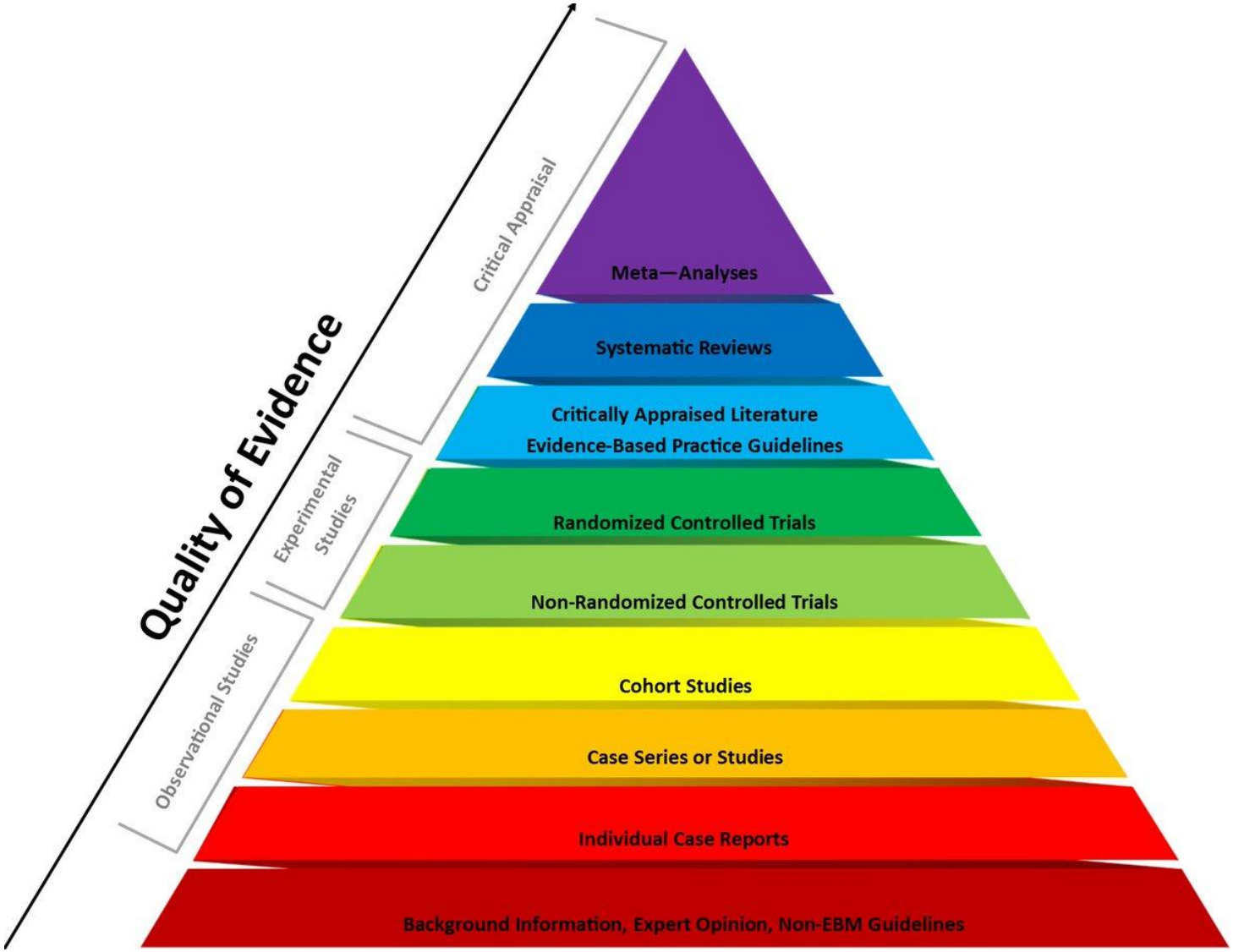


Original and valuable research work

- A journal paper that **synthesizes** the literature in a field.
- Creates a **solid contribution** for all members of the community interested in a particular area or topic.
- Doesn't collect or analyzes **primary data**.
- It **aggregates** and **analyses empirical findings** from the **research work of others**.
- Assess and controls **bias** in **methods** and **recommendations**.
- **Evidence-based practice** concerning **a narrow research question**.
- It may generate new **frameworks** and **theories.**
- It identifies **topics** and **questions** requiring **more investigation**.



Quality of evidence in research



Generic steps of a literature review

Research Question(s)

- Must clearly define the **main objectives**, **concepts** and **variables**.
- It can be **broader** or **narrower** questions, depending on the objective.
- The research questions will **guide all subsequent steps**.

Search Strategy

- Define search **terms**, **queries** and **sources**.
- Can be **exhaustive**: all relevant content, **published/unpublished, grey literature**
- Or **representative**: only top-tier journals, **most influential**

Inclusion Criteria

- Determine the **relevance** of the material found.
- Define rules for deciding **content inclusion and exclusion**.
- Ensure **enhanced objectivity** and **avoid biases or mistakes** (at least 2 reviewers and a procedure to resolve disagreements).

Generic steps of a literature review

Quality Assessment

- Appraise the **rigour** of the **research design and methods** (risk of bias).
- Ascribe **quality scores** or **domain-based evaluations** to determine which studies have been **designed and executed appropriately**.
- **Refine the selection** of content for the final sample.

Data Extraction

- Decide **what data is relevant** to the problem of interest.
- **Gather data** from each study.
- Not just **results**, but also information about **how**, **when**, **where** and **by whom** the study was conducted.

Analysis

- **Collate**, **summarize**, **aggregate**, **organize**, and **compare** the evidence extracted from the included studies.
- **Create knowledge** though a coherent lens that makes sense of the data.
- Identify and discuss **causal effects**.

The tools: types of Literature Reviews

- **Narrative review**
- **Mapping review**
- **Systematic review**
- Umbrella review
- Scoping review
- Integrative review
- Mixed methods review
- …

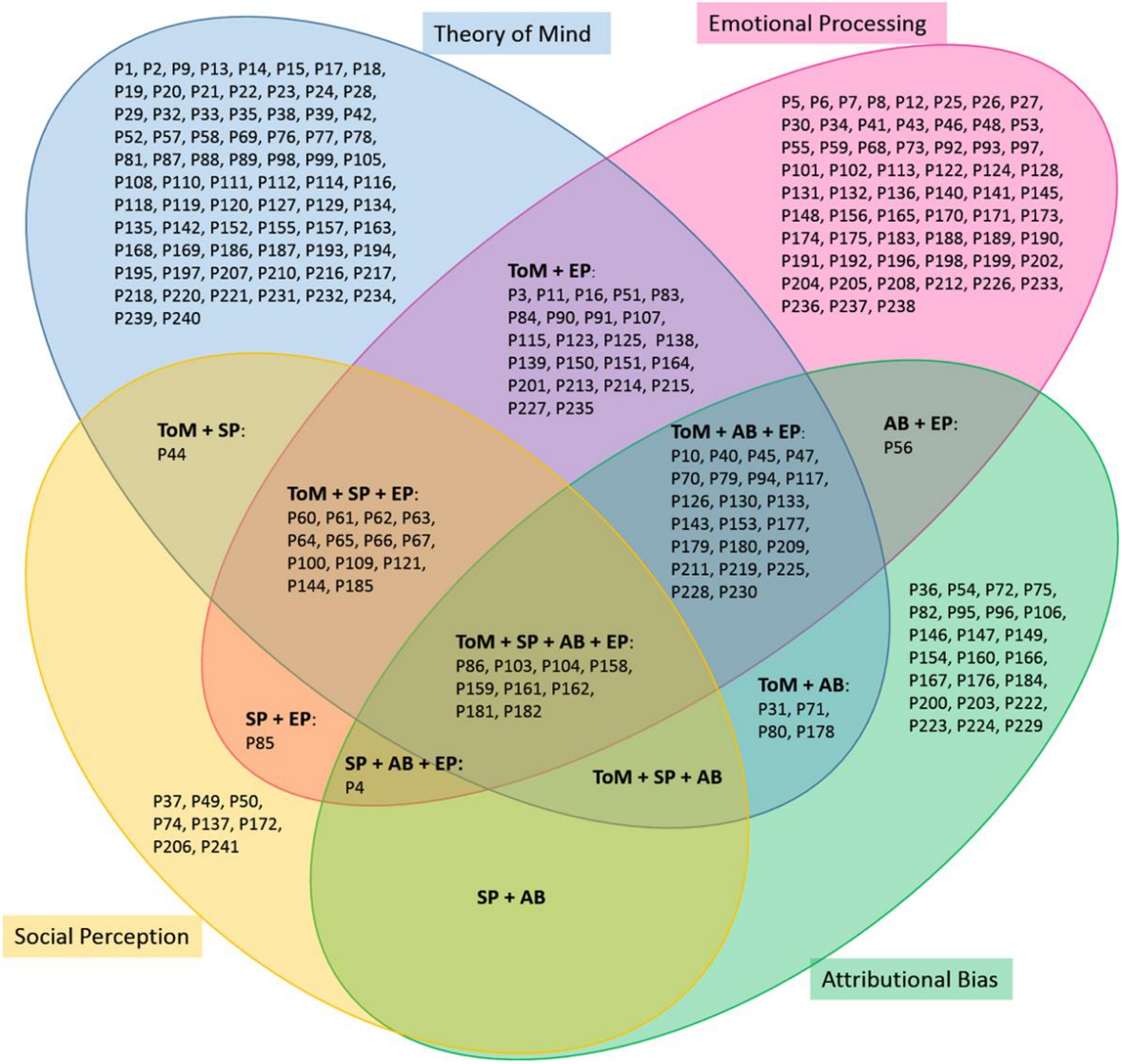
Narrative review

- **Objective**
  - Summarize what has been written on a particular topic.
  - Does not seek generalization or cumulative knowledge from what is reviewed.
- **Search Strategy & Selection Criteria**
  - Selective in nature.
  - Doesn't seek a complete coverage of the literature on the topic of interest.
  - Authors usually select studies that support their own view.
- **Quality Assessment**
  - Authors decide what to include according to the objective.
  - No need to formally assess risk of bias.
- **Analysis**
  - Open narrative.
  - May also follow analysis guidelines: thematic, chronological, conceptual framework, …

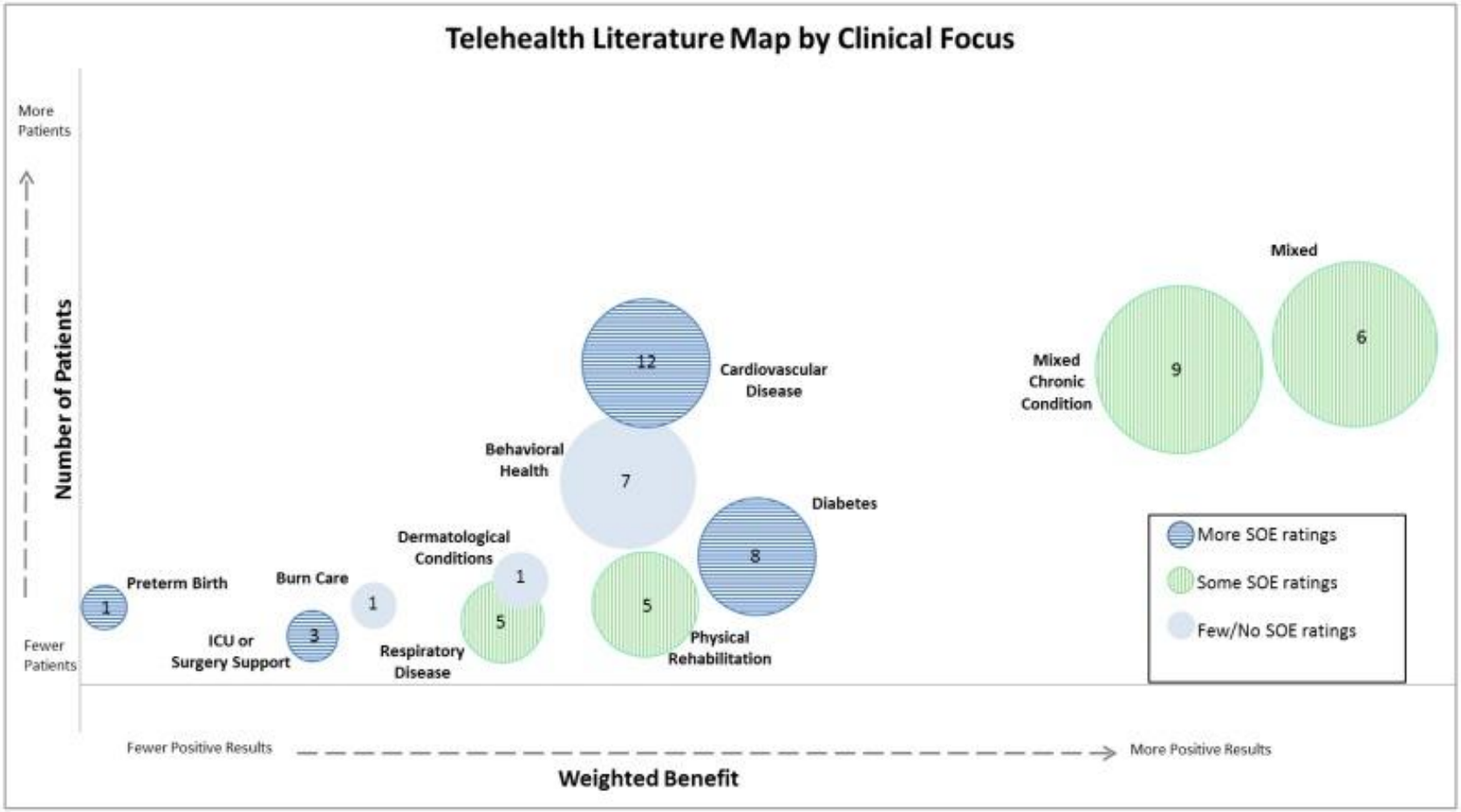
Mapping review

- **Objective**
  - Map (visual synthesis) the literature on a particular topic.
  - Identify gaps in particular area.
  - First step to understand a topic, possibly leading to a systematic review.
- **Search Strategy & Selection Criteria**
  - Comprehensive search.
  - Selection criteria may follow a classification framework (time, theory, effect, evidence quality, …)
- **Quality Assessment**
  - No need to formally assess risk of bias
- **Analysis**
  - Follows a classification framework
  - Graphical presentation of results is preferred

Mapping review



Mapping review



Systematic review

- **Objective**
  - Aggregate, critically appraise, and synthesize in a single source all empirical evidence that meet a set of pre-specified eligibility criteria.
  - Answer a clearly formulated research question in depth to support evidence-based decision-making.
- **Search Strategy & Selection Criteria**
  - Exhaustive literature search, with a priori inclusion and exclusion criteria.
  - Only experimental studies are considered (otherwise it's an integrative review).
- **Quality Assessment**
  - Risk of bias of the studies methods.
  - Quality of evidence by outcome or recommendation.
  - Evaluation is carried following a formal system, e.g. Cochrane criteria and GRADE system.
- **Analysis**
  - Meta-analysis: statistical analysis of study results
  - Qualitative/narrative: comparison tables, content analysis, theoretical frameworks, classification schemes.
  - Review results must be reproducible.

Systematic review question

- A systematic review answers a specific research question.
- There are several frameworks to help create the question.
- PICO
  - Population: who is the focus of my research question?
  - Intervention: what is the proposed, new intervention?
  - Comparison: what is the current or alternative intervention?
  - Outcome: what measurable outcome is affected?
- Example
  - In conscious ICU patients, what is the effect of virtual reality relaxing experiences on delirium compared to rehabilitation therapy?

Other question frameworks

- **PEO**. Useful for qualitative research topics.
  - **P**opulation: who is my question focused on?
  - **E**xposure: what is the issue I am interested in?
  - **O**utcome: what, in relation to the issue, do I want to examine?
- E.g.: What are the daily living experiences of mothers with postnatal depression?
- **SPICE**. Useful for research topics evaluating the outcomes of a service, project, or intervention.
  - **S**etting: Setting is the context for the question (where?).
  - **P**erspective: the users, potential users, or stakeholders of the service (for whom?).
  - **I**ntervention/Interest/Exposure: the action taken for the users, potential users, or stakeholders (what?).
  - **C**omparison: alternative actions or outcomes (compared to what?).
  - **E**valuation: result or measurement that will determine the success of the intervention (what is the result, how well?).
- E.g.: For secondary students in Minho, what is the effect of computing classes once a week on math grades compared to no computing classes?

Systematic review guidelines

- Considering the high level of **rigour** and **reproducibility** of a systematic review, it should always follow well-established **guidelines**.
  - **PRISMA**.
    - Preferred Reporting Items for Systematic Reviews and Meta-Analyses.
  - **IOM**
    - Finding What Works in Health Care: Standards for Systematic Reviews.
- Other important guides
  - Cochrane Handbook for Systematic Reviews of Interventions
  - University of Maryland Libraries' Systematic Review Guide

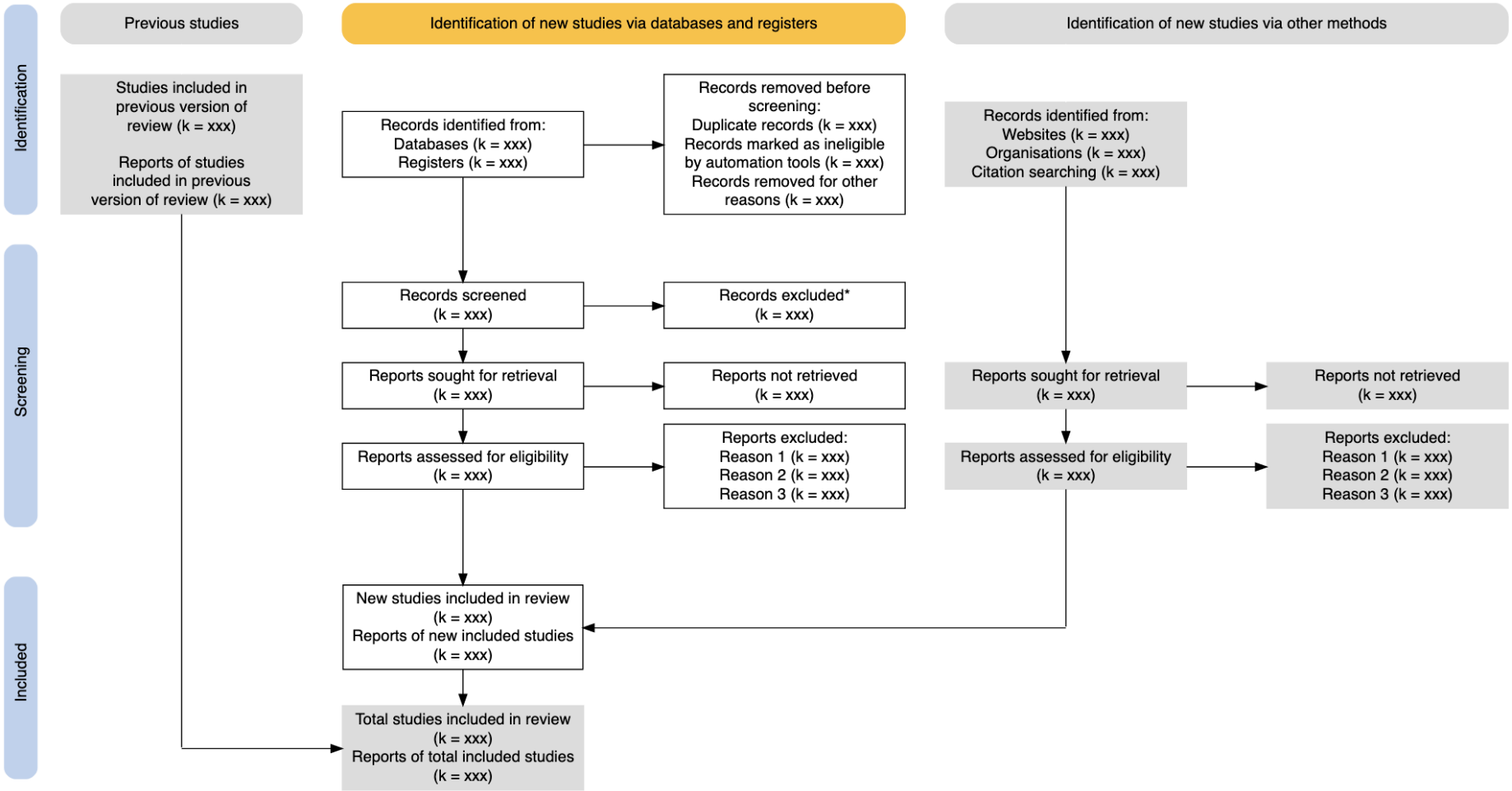
PRISMA Abstract

- Identify the report as a **systematic review**
- Provide an explicit statement of the main **objective**(s) or **question**(s) the review addresses
- Specify the **inclusion** and **exclusion criteria** for the review
- Specify the **information sources** (such as databases, registers) used to identify studies and the date when each was last searched
- Specify the **methods** used to assess **risk of bias** in the included studies
- Specify the **methods** used to present and **synthesise** results
- Give the **total number of included studies** and **participants** and summarise relevant characteristics of studies

PRISMA Abstract

- Present results for **main outcomes**, preferably indicating the number of included studies and participants for each.
  - **If meta-analysis was done**, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (that is, which group is favoured)
- Provide a brief summary of the **limitations of the evidence** included in the review (such as **study risk of bias**, inconsistency, and imprecision)
- Provide a **general interpretation of the results** and important implications
- *Specify the primary source of funding for the review*
- *Provide the register name and registration number*
  - *PROSPERO*
  - *INPLASY*

PRISMA Flow Diagram



* If automation tools were used, indicate how many records were excluded by a human and how many were excluded by automation tools.

Systematic review article

- **Introduction**
  - **Context**. Describe the objective(s) or question(s) context, to familiarize the reader with the research topic.
  - **Question**. Provide an explicit statement of the objective(s) or question(s) the review addresses.
  - **Importance**. Argue about the importance of this review.
  - **Review Type**. Clearly state the type of the review (narrative, mapping, systematic, …)
- **Methods**
  - **Search strategy**. Identify the scientific publication databases and the search query used.
  - **Eligibility criteria**. Clearly define each criterion that articles must meet to be included in the study.
  - **Data Extraction process**. Present the process used to extract data from the articles (reading of titles and abstracts, full reading, number of authors involved in each task, tools used, synthesis methods, …).
  - **Risk of bias**. State how risk of bias was assessed (e.g. Cochrane risk of bias tool).

Systematic review article

- **Results**
  - **Study selection**. Present the (PRISMA) flow diagram showcasing the selection process.
  - **Purpose of Study**. Describe the main objectives of each selected article.
  - **Methodology and study design**. Identify the methodology and research design of each selected article (e.g. quantitative, qualitative, mixed, experimental, survey,…)
  - **Procedures**. Describe the methods of each selected article (e.g. diagnosis methods, technology, platforms, tools, sessions).
  - **Study Design**. Identify the study design of each selected article (e.g. single case, RCT, control groups, placebo usage, number of intervention, study duration, etc)
  - **Assessment Methods**. Describe the evaluation methods used in each selected article (e.g. scales, questionnaires, tests, tools, metrics, instruments, …)
  - **(Statistical) Results**. Summarise the results reported in each selected study.
  - **Risk of Bias Assessment**. Present the results obtained by the risk of bias assessment method described in the in methods section (e.g. Cochrane risk of bias tool).

Systematic review article

- **Discussion**
  - Discussed in the following slides
- **Recommendations for future studies**
  - Not mandatory, but highly recommended
  - The *juice concentrate* of your work
  - Brief and direct to the point considerations for future studies in the area
    - (E.g. assessment methods to be used, data that should be reported, areas in clear need of more investigation, study designs that should be used, technology selection criteria, …)
- **Conclusion**
  - Depends on journal guidelines
  - Typically, a very brief summary of the study objectives, methods and main findings.

Discussion

- The review question is answered here!
- Strangely, this is where we draw conclusions – **produce new knowledge!**
  - From the results collected from the reviewed studies.
  - From external sources: data, theory, other studies, etc.
- The **confidence** of each **conclusion proposition** may vary, from very **strong evidence** to **just a signal**.
- **Convincing empirical evidence and/or deductive arguments** must support each conclusion proposition.

Discussion

- Types of considerations used in the discussion
  - General interpretation of the results in the context of other evidence.
  - Patterns (e.g. conscious ICU patients tolerate well VR)
  - Advance explanations for patterns (e.g. females play less videogames)
  - **Cause-effect processes** (e.g. VR reduces PICS)
  - Limitations in current knowledge about the topic in question
  - Limitations of the review process and studies
  - Lack of knowledge about some important aspect of the topic in question
  - Implications of the results for practice, policy, and future research

Other types of review

- Umbrella review
  - Review of reviews or meta review.
- Scoping review
  - Broad analysis of research literature to provide an **initial indication** of potential size and scope in a topic.
  - Similar to a **mapping review**, but more narrative.
  - Helps to determine the value of undertaking a **full systematic review** and/or to pursue a research area.
- Integrative review
  - Similar to a systematic review but also considers **nonexperimental studies**.
- Mixed methods review
  - Reviews that use both **qualitative** and **quantitative** methods.

Start Reviewing Now!

Literature review presentation

- Each student has **6 minutes** to present the literature review.
- The presentation should include the following content:
  - The problem or gap in the field
  - The research question(s) guiding the review
    - Make sure the research question is appropriate
  - The search query
    - Justify the non-obvious terms used in the query
  - Inclusion/exclusion criteria
    - Include a brief justification for each non-trivial criterium
  - Study selection
    - Include the PRISMA diagram with records identified, records after title/abstract screening, …

Literature review presentation

- Analysis of the Selected Studies
  - **<u>This is the most important content</u>**
  - Examples:
    - "Vibration signals are the most commonly used predictive indicator for bearing failure detection, appearing in over 70% of the reviewed predictive maintenance studies."
    - "SARIMA time-series models achieve the lowest prediction error for short-term hospital admission forecasting, especially when historical admission data is the primary input variable."
- Future Research
  - A clear research question to guide the MSc research plan
  - Evidence-based methods, identified in the literature, to investigate the research question